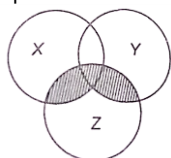




COMPUTER

- The ASCII code of A is
(a) 66D (b) 41H
(c) 01000010 (d) 01100011
- An eight bit byte is capable of representing how many different characters?
(a) 64 (b) 128
(c) 256 (d) 512
- The least significant bit of the binary number, which is equivalent to any odd decimal number is
(a) 0 (b) 1
(c) 1 or 0 (d) All of these
- Which of the following Boolean expression represents the shaded portion of the Venn diagram?
Here, " " represent an AND operation and "+" denotes an OR operation



- $Z' + (X.Y)$
 - $Z.(X + Y)$
 - $(Z.X') + Y$
 - $Z'(X + Y)$
- Consider x and y be some Boolean variables, '+' denotes the OR operation and "." denotes the AND operation. What will be the simplified form of the Boolean expression $x.(x + y)$?
(a) y (b) 1
(c) 0 (d) x
- Which one of the following is not a valid rule of Boolean Algebra?
(a) $A + 1 = 1$ (b) $A = A'$
(c) $AA = A$ (d) $A + 0 = A$
- When two binary numbers are added, then an overflow will never occur, if
(a) both numbers of same sign
(b) the carry into the sign bit position and out of sign bit position are not equal
(c) the carry into the sign bit position and out of sign bit position are equal
(d) the carry into the sign bit position is 1
- The sum of 11010 + 01111 equals to
(a) 101001 (b) 101010
(c) 110101 (d) 101000
- Which protocol needs to be installed for Internet access on a network?
(a) TCP/IP (b) TELNET
(c) IPX/ SPX (d) Net BEUI
- A petabyte represents approximately
(a) 1000 gigabytes (b) 1000 kilobytes
(c) 1000 terabyte (d) 1000 yottabytes

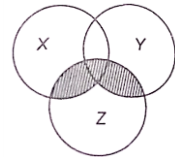
ANSWER

1.	B	2.	C	3.	B	4.	B	5.	D
6.	B	7.	D	8.	A	9.	A	10.	c

SOLUTION

- (b)
The ASCII
Code of A in decimal system is 65. Which can be written as 41 in hexadecimal system

16	65	1
	4	

 $\Rightarrow (65)_{10} = (41)_{16} = 41H$
- (c)
We know that, n bit byte can represent $(2)^n$ different characters. Hence 8 bit byte can represent $(2)^8 = 256$ characters.
- (b)
Least significant bit is the only bit which is equal to odd number in magnitude ($2^0 = 1$).
It represent any odd decimal number.
This bit must be included. So, in any an odd decimal number least significant bit must be 1.
- (b) Given Venn-diagram
In set Theory, this Venn-diagram shows the following expression.

 $Z \cap (X \cup Y)$
We know that, in Boolean Algebra
 $\cap$ = AND operation = $\cup$ = OR operation = +
From Eq. (i), the required Boolean expression is $Z.(X + Y)$
- (d) The given Boolean expression is
Let $E = x.(x+y)$
 $= x.x + x.y$ "By distributive law"
 $= x + x.y$ "By idempotent law"
 $= x.(1+y)$ "By boundness law"
 $= x$ "By identity law"
Which is the required simplified form of E.
- (b) From options
(i) $A + 1 = 1$ is boundness law
(ii) $AA = A$ is idempotent law
(iii) $A + 0 = A$ is identity law
(iv) $A = (A)'$ is involution law
But $A = A'$ is not a valid rule of Boolean algebra which is valid in matrix and is called symmetric matrix.
- (d) When two binary numbers are added, then an overflow will never occur if, the carry into the sign bit position is 1.
- (a)

$$\begin{array}{r} 11010 \\ + 01111 \\ \hline 101001 \end{array}$$

- (a) TCP/IP $\Rightarrow$ Transmission Control Protocol/ Internet Protocol.

- (c) We know that, 1 Petabyte = 2^{50} bytes.
 $= 10^{15} = 1000$ Terabyte
 $(\because \text{Terabyte} = 2^{40} \approx 10^{12})$

